MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PEC-IT601B Distributed Systems

UPID : 006590

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) The node where the distributed transaction originates is called the .....

(II) What is the advantage of a DDBMS?

(III) What is view management in distributed database design?

(IV) What are the factors that govern query optimization?

(V) What are distributed concurrency control algorithms?

(VI) What is shared-memory architecture?

(VII) What is the purpose of mobile database replication?

(VIII) ..... protocol is used to perform multiple transactions that execute on different database.

(IX) What is fragmentation in distributed database design?

(X) What are the goals of transaction management?

(XI) What are the parallel architectures used in parallel database systems?

(XII) What is distributed object management?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. What is concurrency control? What is lock based protocol? [5]
3. 1) What is distributed transparency?Explain different levels of distributed transparency(level 1,level 2,level 3) with a diagram. [5]
4. Write short note on mobile database. [5]
5. What are the advantage and disadvantage of replication? What is auxiliary program? [5]
6. Write short note on MDBS architecture. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Discuss various transaction model, [5]
   (b) What is flat and nested transaction? Explain with example. [5]
   (c) What is serializability in distributed database? [5]
8. (a) Why is it necessary to have a commit protocol? [5]
   (b) Explain how does two phase commit protocol respond to failure of a participating site. [5]
   (c) What are the disadvantages of 2 phase commit protocol? [5]
9. (a) Explain different parallel database architecture. [7]
   (b) What is the goal of using parallel database architecture. [4]
   (c) Difference between parallel database and distributed database. [4]
10. (a) What is serializability in distributed database? [5]
    (b) Write down the algorithms both Coordinator and Par ticipants of 2PC protocol in distributed environment. [5]
    (c) Explain with diagram communication structure for different 2PC protocol. [5]
11. (a) Discuss multidatabase architecture. [5]
    (b) What is mobile database? [5]
    (c) Discuss in brief concurrency control mechanism in distributed DBMS. [5]

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